

Chapter 1: Physical Quantities & Measurement

Key Concepts

Physical Quantity — Anything that can be measured (length, mass, time, etc.)

Base Quantities — 7 fundamental quantities that cannot be derived:

- Length (meter, m), Mass (kilogram, kg), Time (second, s)
- Electric Current (ampere, A), Temperature (kelvin, K)
- Amount of Substance (mole, mol), Luminous Intensity (candela, cd)

Derived Quantities — Formed from base quantities (speed = length/time, density = mass/volume)

Scientific Notation — Express very large/small numbers as $a \times 10^b$ where $1 \leq a < 10$

Prefixes — kilo (k, 10^3), mega (M, 10^6), giga (G, 10^9), milli (m, 10^{-3}), micro (μ , 10^{-6}), nano (n, 10^{-9}), pico (p, 10^{-12})

Significant Figures — All non-zero digits count. Zeros between non-zero digits count. Leading zeros don't. Trailing zeros with decimal count.

Measuring Instruments — Least count

- Vernier caliper — 0.1 mm (0.01 cm) least count
- Screw gauge — 0.01 mm (0.001 cm) least count
- Stopwatch — 0.1 s or 0.01 s
- Physical balance / Electronic balance

Zero Error — When an instrument shows non-zero reading with nothing measured. Must be added or subtracted.

Precision & Accuracy — Precision = closeness of repeated measurements. Accuracy = closeness to true value.

MCQ's

1. Which is a base quantity?

A. Speed B. Force C. Length D. Density

Answer: C

2. SI unit of electric current is:

A. Volt B. Ampere C. Ohm D. Coulomb

Answer: B

3. 1 kilometer = ?

A. 10^2 m B. 10^3 m C. 10^4 m D. 10^5 m

Answer: B

4. Least count of vernier caliper is:

A. 1 mm B. 0.1 mm C. 0.01 mm D. 0.001 mm

Answer: B

5. Number of significant figures in 0.00560 is:

A. 2 B. 3 C. 4 D. 5

Answer: B (5, 6, and trailing zero with decimal = 3)

6. The SI unit of force is:

A. Newton B. Joule C. Watt D. Pascal

Answer: A

7. 1 micrometer = ?

A. 10^{-3} m B. 10^{-4} m C. 10^{-5} m D. 10^{-6} m

Answer: B

8. Which of these is not a derived quantity?

A. Speed B. Volume C. Mass D. Density

Answer: C

9. Least count of screw gauge is:

A. 0.1 mm B. 0.01 mm C. 0.001 mm D. 0.0001 mm

Ans:

- 100.0 g has 4 significant figures

- [illegible]

Q1. Convert the following:

- ### Solutions:

- [illegible]

(a) 4500000
(b) 0.00034
(c) 0.006700

Solutions:

- [illegible]

(a) 0.04050
(b) 5600
(c) 203.00
(d) 0.0008

Solutions:

- [illegible]

Solution:

[illegible]

Solution:

[illegible]

Solution:

$$\text{Area} = 5.2 \times 3.8 = 19.76 \text{ cm}^2$$

Since both measurements have 2 significant figures, the answer should have 2 significant figures.

Area = 20 cm²

[illegible]

Q7. A car travels 150 m in 5.0 seconds. Find its average speed.

Solution:

$$\text{Speed} = \text{Distance} / \text{Time} = 150 / 5.0 = 30 \text{ m/s}$$

[illegible]

Q8. The mass of a metal block is 245.6 g and its volume is 25.0 cm³. Calculate its density.

Solution:

$$\text{Density} = \text{Mass} / \text{Volume} = 245.6 / 25.0 = 9.824 \text{ g/cm}^3$$

Mass has 4 significant figures, volume has 3. Answer should have 3.

Density = 9.82 g/cm³